Mid term physics

1a) force on $X = 8$ cm if $q = 1.00\mu c$? $+q = 3cm$ $-2q = 8cm$ $+q = 11cm$

$F = K\frac{|q_1 q_2|}{r^2}$

$+q = 3cm$

$r = 8-3 = 5 = 0.05m$

$F_1 = \frac{k|(1\times10^{-6})(-2\times10^{-6})|}{0.05^2}$

$= (9\times10^9)\frac{2\times10^{-12}}{0.0025}$

$= 7.20N$ (left)

$+q = 11cm$

$r = 11-8 = 0.03m$

$F_2 = \frac{k|(1\times10^{-6})(-2\times10^{-6})|}{0.03^2}$

$= (9\times10^9)(2.22\times10^{-9})$

$= 20.0$ N (right)

$F_{net} = 20 - 7.2 = 12.80$ N Right

b) $x = 11cm$, $-2q = 1cm$, $+q = 5 + 3q = 8$, $-q = 14$ $K = 9\times10^9$

$q = 1\mu c$

$E = k\frac{|q|}{r^2}$

$+q = 5cm$

$r = 0.06$ m

$E_2 = (9\times10^9)\frac{1\times10^{-6}}{0.0036}$

$E_2 = 2500$ NC (right)

$+3q - 8cm$

$r = 0.03m$

$E_3 = k(\frac{3\times10^{-6}}{.0009})$

$E_3 = 30,000$ N/C (right)

$-2q = 1cm$

$r = 0.1m$

$E_1 = k\frac{2\times10^{-6}}{(.1)^2}$

$E_1 = 1800$ N/c (left)

$-q = 14cm$

$r = 0.03$ m

$E_4 = k(\frac{1\times10^{-6}}{0.0009})$

$E_4 = 10,000$ N/c (left)

$E_{net} = -1800 + 2500 + 30,000 - 10,000$

$E_{net} = 20700$ N/c

$E_{net} = 2.07\times10^4$ N/c

2a)

$q_a = q_b = +7.5\ \mu c$ force = x

$q_c = q_d = -7.5\mu c$

Since $q_a = q_b + q_c = q_d$ the net force on $q = 0$ N

b $q = 2\mu c$, $l = 0.1$ m

$K = 9\times10^9$

distance from point to q = $r = \frac{s}{\sqrt{2}} = \frac{0.10}{\sqrt{2}} = 0.0707$ m

$F = k\frac{|qQ|}{r^2} = k\frac{(2\times10^{-6})(7.5\times10^{-6})}{.0707^2} = (9\times10^9)(3\times10^{-9}) = 27.0$ N, the force on q is still 0N.

3) 40 kV

non-relativistically find the max speed

$qV = \frac{1}{2}mv^2 \rightarrow v = \sqrt{\frac{2qV}{m}}$

$q = 1.6\times10^{-19}$ C $V = 40,000 = 4\times10^4$ $m = 9.11\times10^{-31}$ kg $v = \sqrt{\frac{2qV}{m}}$ $V = \sqrt{\frac{2(1.6\times10^{-19})(4\times10^4)}{9.11\times10^{-31}}}$

$v = 1.19\times10^8\ \frac{m}{s}$

4) a) 4 cm separation : $7.5\times10^4\ Vm^{-1}$

what's ΔV btwn plates?

$\Delta V = E\cdot d = 7.5\times10^4(0.04m)$

$\Delta V = 3.0\times10^3 V$

b) 1cm from lower

$V = 7.5\times10^4(0.01m)$

$V = 750$ V

c) $E = \frac{\Delta V}{d}$ $\Delta V = 80$ mV

$d = 9nm$

$E = \frac{80\times10^{-3}}{9\times10^{-9}} = 8.89\times10^6$ V/m

5) double charged ion $(E) = 32.0$ KeV, separated by 2cm

$E = 32$ kev $= 32\times10^3\times1.602\times10^{-19}$ J $\rightarrow q = 2e = 2\times1.602\times10^{-19}\times2 = 3.204\times10^{-19}$ C

$\Delta V = \frac{E}{q} = \frac{32\times10^3\times1.602\times10^{-19}}{3.204\times10^{-19}} \Rightarrow \Delta V = 16,000\ V / 0.02$ m $= 8.00\times10^5\ \frac{V}{m}$

4) energy of a nucleus was 5 MeV, how close to 79 protons before deflection?

$U(r) = \frac{K \cdot Z_{gold} \cdot Z_{\alpha} \cdot e^2}{r}$ $\quad E = 5\,MeV = 5\times10^6\,eV$

$r = \frac{K \cdot Z_{gold} + Z_{\alpha} \cdot e^2}{E}$ $\quad Z_{gold} = 79$ protons $Z_{\alpha} = 2$, $K = 8.99\times10^9$ $\quad e = 1.602\times10^{-19}\,C$

$R = \frac{(8.99\times10^9)(79)(2)(1.602\times10^{-19})^2}{(5\times10^6)(1.602\times10^{-19})} = \frac{(1.846\times10^{-28})}{8.01\times10^{-13}} = 2.3\times10^{-13}\,m$

Unit 1

1) Capacitance to store $3\,\mu C$ at $V = 120$? $\quad C = \frac{Q}{V} = \frac{3\cdot10^{-6}\,C}{120\,V} = 2.5\times10^{-8}\,F$

2 a) energy = ? $10\,\mu F$ capacitor to $9\times10^3\,V$

$E = \frac{1}{2} CV^2$

$E = \frac{1}{2}(10\times10^{-6})(9\times10^3)^2 = \frac{1}{2}(10\times10^{-6})(8.1\times10^7)$

$E = 4.05\times10^2\,J$

$E = 405\,J$

2b) $8\,\mu F$ that stores $40\,J$

$E = \frac{1}{2}CV^2$ $\quad 8\,\mu F = 8\times10^{-6}\,F$

$V = \sqrt{\frac{2E}{C}}$

$V = \sqrt{\frac{2(40)}{8\times10^{-6}}}$

$V = \sqrt{\frac{80}{8\times10^{-6}}}$

$V = \sqrt{1\times10^7}$

$V = 3162.3\,V$

3a) 4 capacitors in parallel

$E = 405\,J$, $V = 9\times10^3$

$C_{tot} = C_1 + C_2 + C_3 + C_4 = 4c$

$E = \frac{1}{2}(4c)V^2$

$405 = 2c(9\times10^3)^2$

$405 = 2c(8.1\times10^7)$

$405 = 1.62\times10^8 C$

$C = \frac{405}{1.62\times10^8} = 2.5\times10^{-6}\,F = 2.5\,\mu F$

3b) what about in series? why n

each capacitor: $Q = 2.5\times10^{-6} \times 9\times10^3$ $\quad Q = 22.5\,mC$

$\frac{1}{C_{tot}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \frac{1}{C_4}$

$= \frac{1}{2.5} + \frac{1}{2.5} + \frac{1}{2.5} + \frac{1}{2.5}$ vs. $2.5 + 2.5 + 2.5 + 2.5$

Parallel has more charge and thats why the designer wants them like so!

4. $1\,k\Omega$ resistor $3V$ battery

a) flow from battery, is $3\,\Omega$ in the led resistor?

$R_{led} = 3\,\Omega$, $R_{resist} = 1\,k\Omega \Rightarrow 1000\,\Omega$, $V_{tot} = 3V$

$I = \frac{V}{R_{tot}} \Rightarrow \frac{3V}{1003\,\Omega} = 0.00299\,A \Rightarrow 299\,mA$

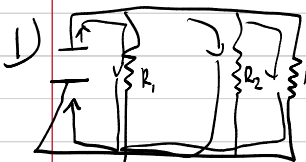
b) Power consumed by LED?

$P = I^2 \cdot R_{led}$

$P = (0.00299\,A)^2 \cdot 3\,\Omega$

$P = 8.94\times10^{-6}\,A\,(3\,\Omega) \Rightarrow P = 2.68\times10^{-5}\,W = 26.8\,\mu W$

Unit 2

1) 

$R_1 = 10k\Omega$ $R_2 = 5k\Omega$ $R_3 = ?$

$R_{tot} = 2k\Omega$
$= 2000\,\Omega$

a) Resistance of R_3?

in parallel : $\frac{1}{R_{tot}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$$\frac{1}{2000} = \frac{1}{10000} + \frac{1}{5000} + \frac{1}{R_3}$$

$$\frac{1}{2000} = \frac{3}{10{,}000} + \frac{1}{R_3} \Rightarrow \frac{1}{R_3} = \left(\frac{1}{2000}\right)^{\times 3} = \frac{3}{10{,}000}$$

$$\frac{1}{R_3} = \frac{5-3}{10{,}000} \qquad R_3 = \frac{10000}{2} = \frac{5000\,\Omega}{5k\Omega}$$

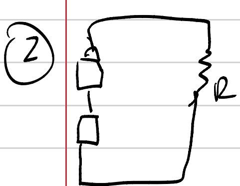
b) $R_{tot} = 2000\,\Omega$ $\Delta V = 12V$

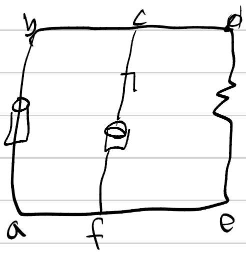
$I = \frac{V}{R_{tot}} \Rightarrow I = \frac{12V}{2000\Omega} = \frac{0.006A}{6mA}$

c) $I_1 = \frac{12V}{10000\,\Omega} = 0.0012\ A = 1.2\ mA$

$I_2 = \frac{12V}{5000\,\Omega} = 0.0024\ A = 2.4\ mA$

$I_3 = \frac{12V}{5000\Omega} = 2.4\ mA$

(2) 



$R = 0.5\Omega$

3 1.58V resistance of $0.02\,\Omega$ in series
w/ 1.53 V carbon-zinc w 0.1 Ω resistance
load res. 10 Ω

a) current flows? $I = \frac{V}{R}$

$$I = \frac{(3 \times 1.58V) + 1.53V}{(3 \times 0.02\,\Omega) + 10.1\,\Omega} = \frac{6.27V}{10.16\,\Omega} = 0.617A \text{ or } 617\ mA$$

b) power to load?

$P = I^2R = (0.617)^2 \times 10\,\Omega = \boxed{3.8\ W}$

c) Dry cell = had only 0.5W to load

$I = \sqrt{\frac{P}{R}} = \sqrt{\frac{0.5}{10}} = \sqrt{0.05} = 0.224A$

$R_{tot} = \frac{V_{tot}}{I} = \frac{6.27\ V}{0.224A} = 28.04\,\Omega$

$R_{internal} = R_{tot} - R_{load} = 28.04\,\Omega - 10\,\Omega = 18.04\,\Omega$

2) $R = 0.5\ k\Omega = 500\,\Omega$

a) $V_{series} = 1.5 + 1.5 = 3V$

$V_{parallel} = 3V$

Series:

$I_{series} = \frac{V}{R} = \frac{3V}{500\Omega} = 0.006A$ ← current = 6.0 mA

$P_{series} = I^2R = (.006)^2(500) = 0.018W = 18mW$ power

Parrallel : $I_{parallel} = \frac{3V}{500\,\Omega} = 6.0\ mA$ current

$P_{parrallel} = (0.006)^2(500) = 18\ mW$ Power

b) $r = 0.5\,\Omega$

Series
$V = 3V,\ R = 1\,\Omega,\ R_{tot} = 501\,\Omega$
$I = \frac{V}{R} = \frac{3V}{501\,\Omega} = 5.99$ mA current

$P = I^2R = 5.99^2(500) = 17.94$ mW power
$P = I^2R = 5.995^2(500) = 17.97$ mW power

parralel
$0.5/2 = 0.25 \quad R_{tot} = 500.25\,\Omega$
$I = \frac{3V}{500.25} = 5.995$ mA current

Unit 3

1)

case	V-direction	B direction	F direction
(a)	$\hat{i}$, rt	$\hat{j}$, up	$-\hat{k}$ (into page)
(b)	$\hat{j}$, up	$\hat{k}$, out page	$\hat{i}$, left
(c)	$\hat{k}$, out page	$\hat{i}$, is right	$\hat{j}$, down

$F = q\,|v \times B|$
negative = $q < 0$

a) $\hat{i} \times \hat{j} = \hat{k}$
v is right $\hat{i}$
B is $\hat{j}$ up, $q < 0$ so $-k$ into page

b) $\hat{j} \times \hat{k} = \hat{i}$
v is $\hat{j}$ up, $\hat{k}$ is out
$-\hat{i}$ is left

c) $\hat{k} \times \hat{i} = \hat{j}$
$\hat{k}$ is out page
$\hat{i}$ is right, $\hat{j}$ is ↓

2) 4×10^3 m/s $= V$, V, $1.25\,T = B$, $F = 1.4 \times 10^{-16}$ N $q = 1.6 \times 10^{-19}$ C

$F = qvB\sin\theta$
$\sin\theta = \frac{F}{qvB}$

$\sin\theta = \frac{1.4 \times 10^{-16}}{(1.6 \times 10^{-19})(4 \times 10^3)(1.25)} = \frac{1.4 \times 10^{-16}}{8 \times 10^{-16}} = 0.175$

Case A
$-\hat{j}$ (left)

$\theta = \arcsin(.175)$
$\theta = 10.1°$ or
$180 - 10.1 = 169.9°$

3a) O-16 ion mass $= 2.66 \times 10^{-22}$ kg at 5×10^6 m/s
Prepare to 1.2 T, circle are 0.231 m radius
what positive charge is anion?

$qvB = \frac{mv^2}{r} \Rightarrow q = \frac{mv}{Br}$

$q = \frac{(2.66 \times 10^{-26})(5 \times 10^6)}{(1.2)(0.231)}$

$q = \frac{1.33 \times 10^{-19}}{0.2772} = \boxed{4.8 \times 10^{-19}\,C}$

(b) Ratio of charge to electron?
$\frac{q}{e} = \frac{4.8 \times 10^{-19}}{1.6 \times 10^{-19}} \rightarrow 3 \Rightarrow$ 3 times the charge of an electron

c) Radius for Oxygen-18?
Scaling so: $\frac{r_{18}}{r_{16}} = \frac{m_{18}}{m_{16}} = 1.125$

$r_{18} = 1.125 \times r_{16}$
$R_0 = 1.125 \times 0.231 = 0.26$ m